PRELIMINARY — NOT APPROVED FOR FABRICATION OR ENERGIZATION

PROJECT BUTTON HR-V0 ELECTRICAL V3 CONNECTED CANDIDATE

Separate RESET and ARM, two PNOZ stages, dual watchdog contacts, external adapters, redundant actuator interruption.

01 External listed sources and DC boundaries
File: 01_external_sources.kicad_sch

02 Dual-channel E-stop and RESET eligibility
File: 02_estop_eligibility.kicad_sch

03 Distinct ARM and watchdog eligibility
File: 03_arm_watchdog_eligibility.kicad_sch

04 Contactor coils, mirror contacts and EDM
File: 04_contactor_edm.kicad_sch

05 Redundant actuator-power interruption
File: 05_actuator_interruption.kicad_sch

06 Protected actuator branches and central DYNAMIXEL star injection
File: 06_branches_and_injection.kicad_sch

07 Independent watchdog power, controller and drivers
File: 07_watchdog_control.kicad_sch

08 Calculated dual-channel 24 V watchdog feedback
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09 Compute, debug and control terminals
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10 U2D2, actuator ports and bonding boundary
File: 10_actuator_interfaces.kicad_sch

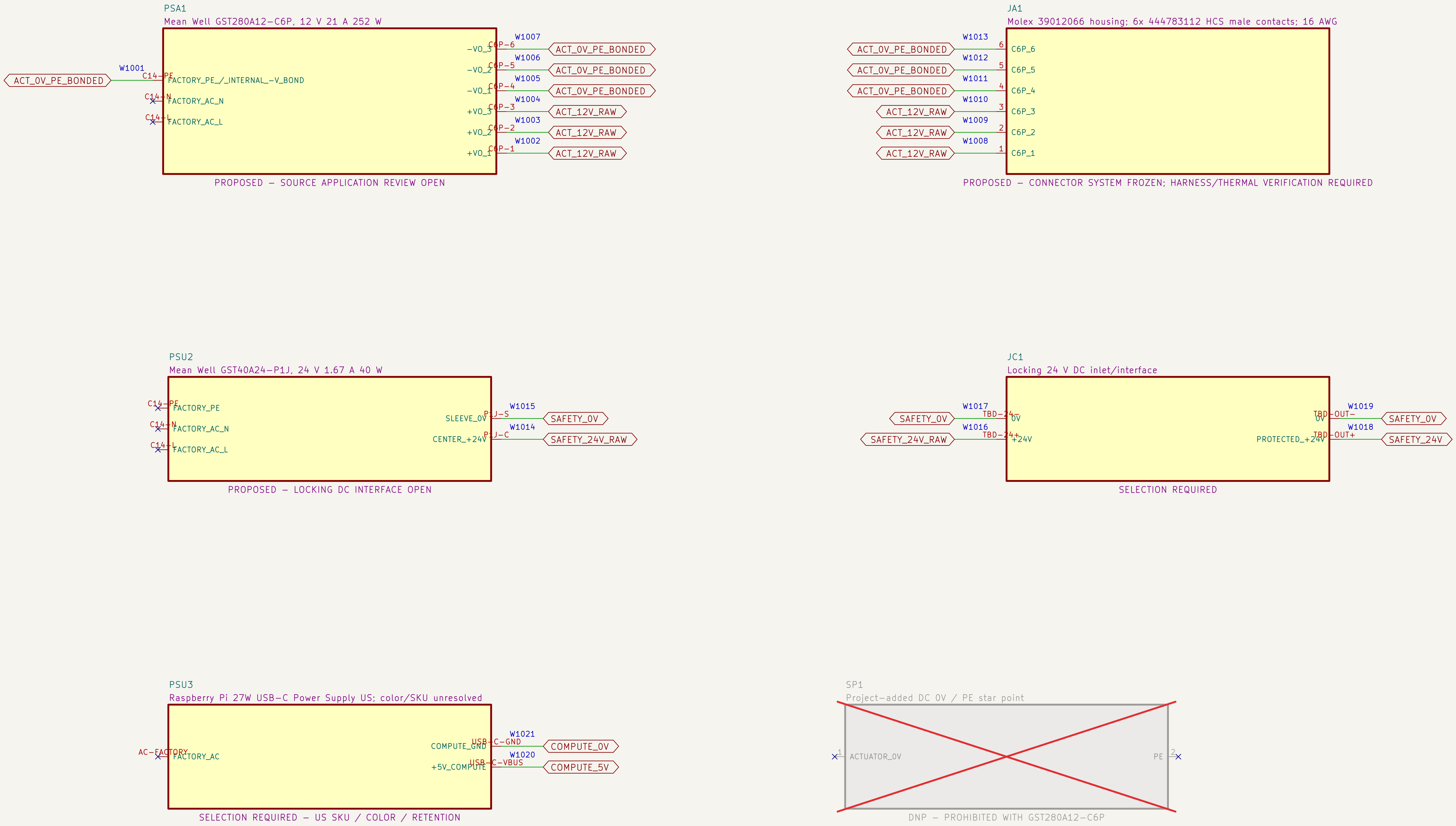
11 Watchdog PCB external connectors
File: 11_watchdog_pcb_connectors.kicad_sch

12 Watchdog PCB test access
File: 12_watchdog_pcb_test_access.kicad_sch

V2.1 PRESERVED; V3 IS A CONNECTED CANDIDATE
PRELIMINARY — NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /
File: project-button-v3.kicad_sch
Title: PROJECT BUTTON HR-V0 ELECTRICAL V3 CONNECTED CANDIDATE index
Size: A3 | Date: 2026-08-07 | **Rev: V3-P1.4**
KiCad E.D.A. 10.0.5 | Id: 1/13

01 External listed sources and DC boundaries

Factory–sealed adapters eliminate project–built mains wiring.



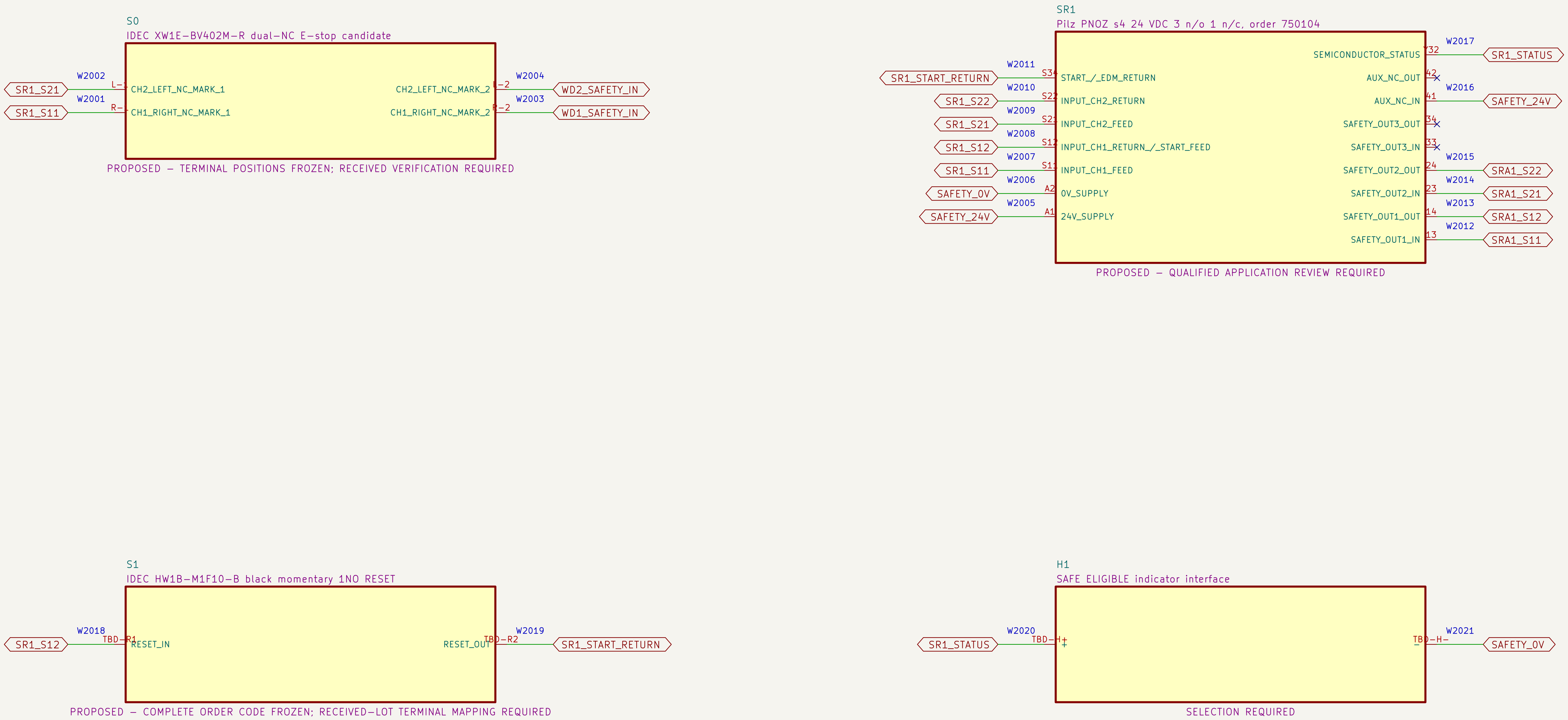
NOTE 1: All AC conductors shown are factory boundaries, not project–built wiring.

NOTE 2: Site cords, receptacles, GFCI/code basis and source application review remain open.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /01 External listed sources and DC boundaries/
File: 01_external_sources.kicad_sch
Title: PB HR–V0 V3–P1.4 – 01
Size: A3 Date: 2026–08–07 **Rev: V3–P1.4**
KiCad E.D.A. 10.0.5 Id: 2/13

02 Dual-channel E-stop and RESET eligibility

Each SR1 input return contains one E-stop NC and one watchdog NO contact; RESET cannot energize K1/K2.



NOTE 1: Heartbeat loss opens both SR1 input returns; recovery alone cannot restore the monitored RESET stage.

NOTE 2: RESET release may make SR1 eligible, but SRA1 and K1/K2 remain de-energized until a later ARM.

CONNECTED DESIGN CANDIDATE — QUALIFIED REVIEW REQUIRED
PRELIMINARY — NOT APPROVED FOR FABRICATION OR ENERGIZATION

Project Button

Sheet: /02 Dual-channel E-stop and RESET eligibility/
File: 02_estop_eligibility.kicad_sch

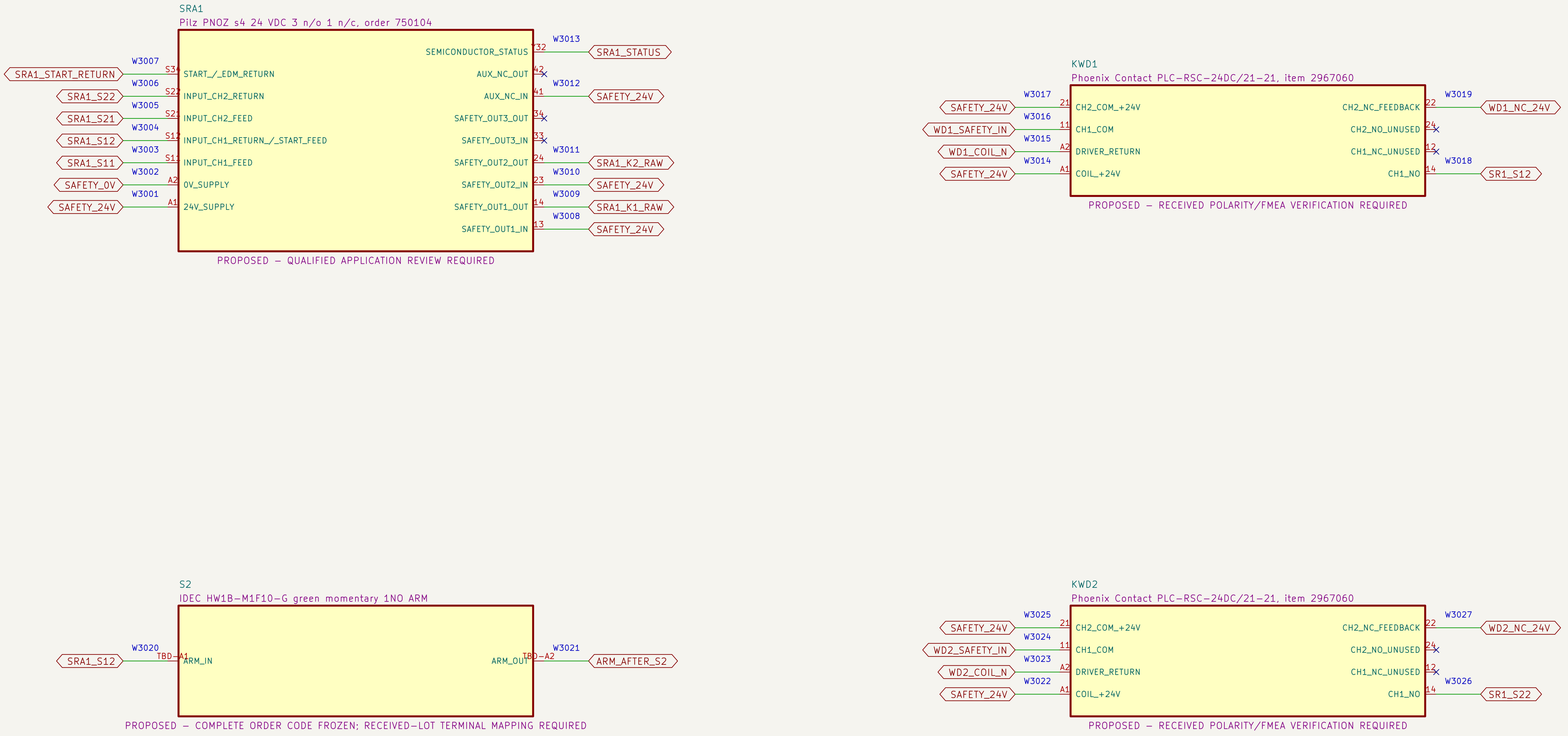
Title: PB HR-V0 V3-P1.4 — 02

Size: A3	Date: 2026-08-07	**Rev: V3-P1.4**
KiCad E.D.A. 10.0.5		Id: 3/13

PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION

03 Distinct ARM and watchdog eligibility

SRA1 requires SR1 eligibility, two watchdog channels, EDM proof and a new ARM action.



NOTE 1: Required after E–stop/watchdog dropout: cause healthy –> RESET press/release –>
SAFE_READY –> distinct ARM press/release.

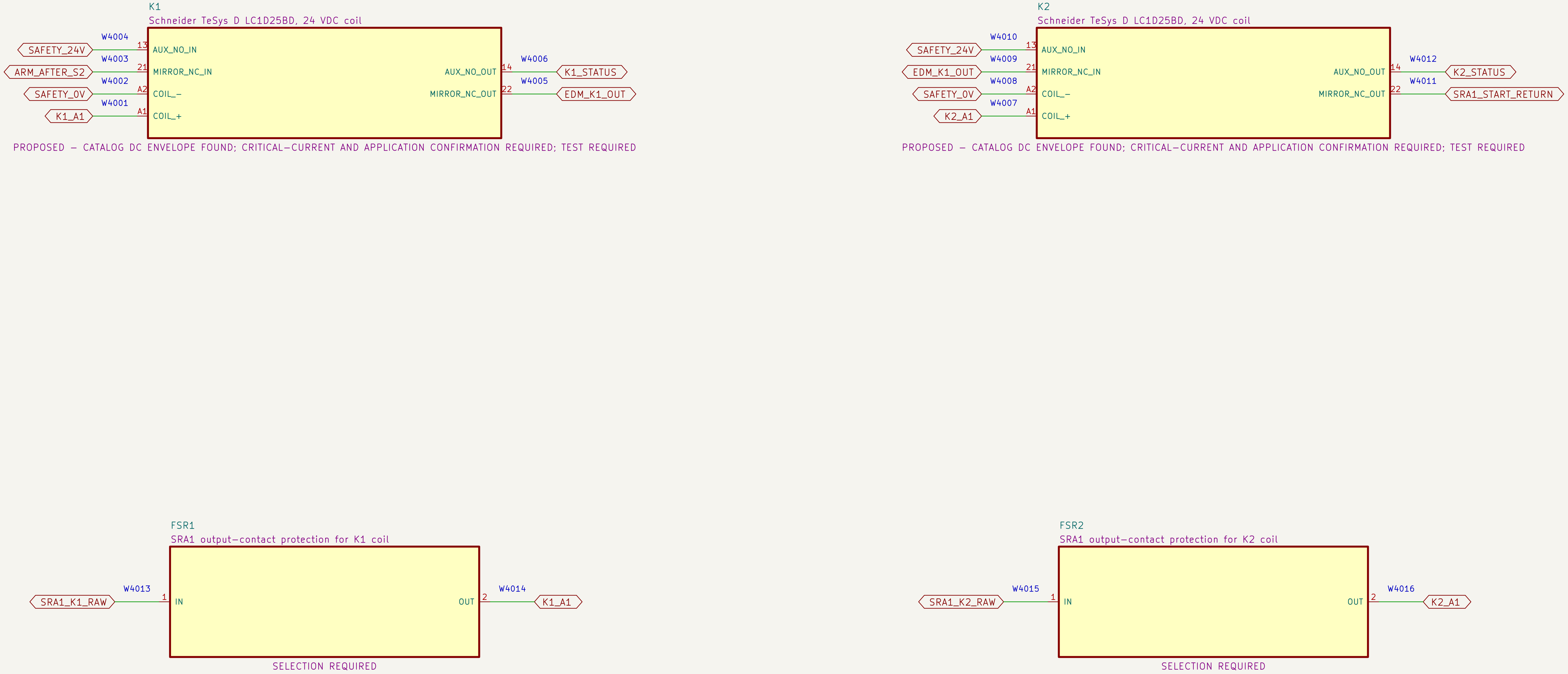
NOTE 2: KWD contacts are in the SR1 returns; SRA1 receives the two SR1 safety outputs directly.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /03 Distinct ARM and watchdog eligibility/
File: 03_arm_watchdog_eligibility.kicad_sch
Title: PB HR–V0 V3–P1.4 – 03

Size: A3	Date: 2026–08–07	**Rev: V3–P1.4**
KiCad E.D.A. 10.0.5		Id: 4/13

04 Contactor coils, mirror contacts and EDM

K1 and K2 are distinct final elements; their mirror contacts form the monitored restart return.



NOTE 1: SRA1 outputs are separately protected before K1 and K2 coils; exact protection remains selection required.

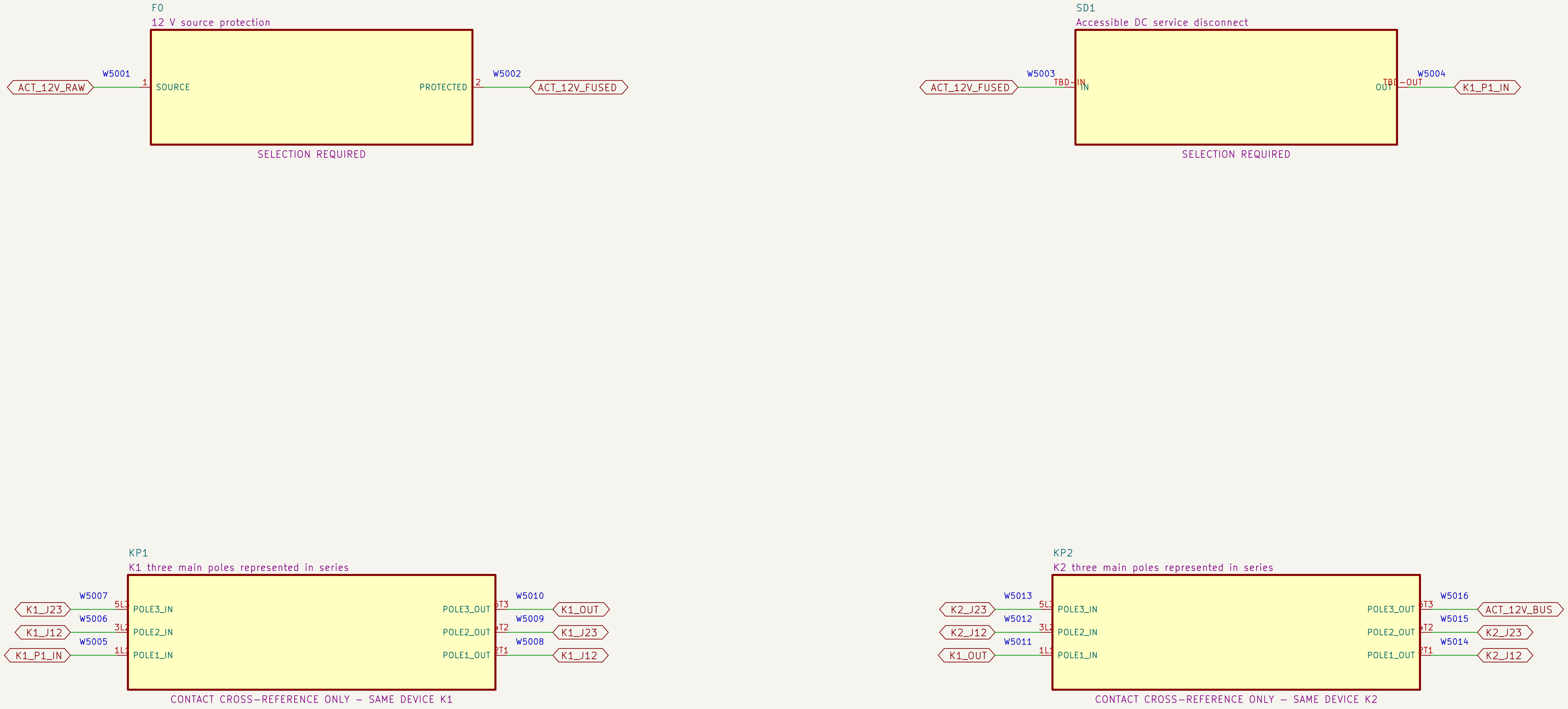
NOTE 2: Loaded DC interruption, suppression behavior, mirror–contact use and coordination require qualified review.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /04 Contactor coils, mirror contacts and EDM/
File: 04_contactor_edm.kicad_sch
Title: PB HR–V0 V3–P1.4 – 04
Size: A3 | Date: 2026-08-07 | **Rev: V3–P1.4**
KiCad E.D.A. 10.0.5 | Id: 5/13

PRELIMINARY — NOT APPROVED FOR FABRICATION OR ENERGIZATION

05 Redundant actuator—power interruption

Source protection, service disconnect and all three poles of K1 then K2 are represented in series.



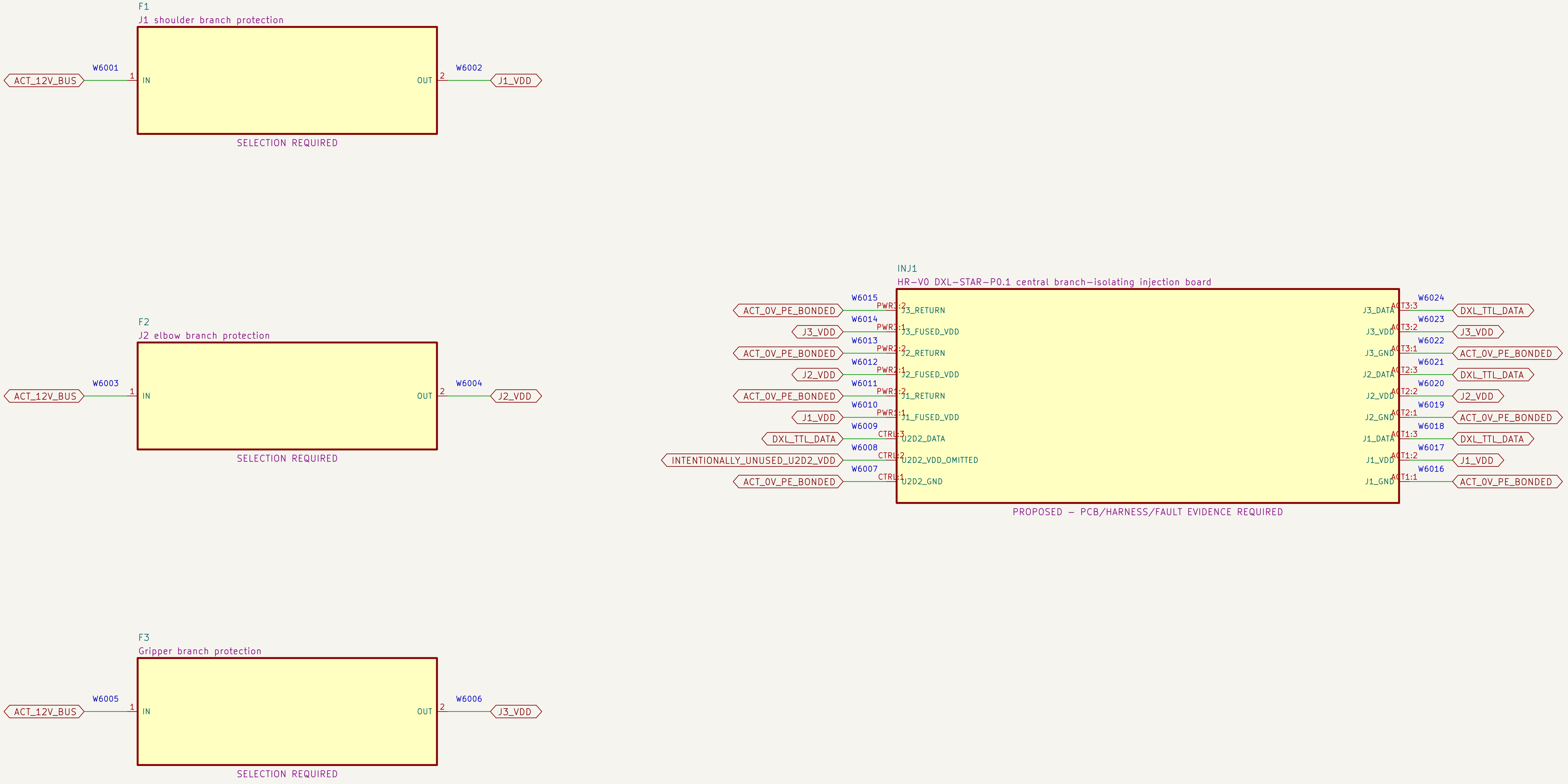
NOTE 1: Series jumpers: K1 2T1->3L2, 4T2->5L3, 6T3->K2 1L1; repeat through K2 to ACT_12V_BUS.

NOTE 2: No fuse, disconnect, conductor, connector or contactor application is released without fault-current evidence.

CONNECTED DESIGN CANDIDATE — QUALIFIED REVIEW REQUIRED
PRELIMINARY — NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /05 Redundant actuator-power interruption/
File: 05_actuator_interruption.kicad_sch
Title: PB HR-V0 V3-P1.4 — 05
Size: A3 | Date: 2026-08-07 | **Rev: V3-P1.4**
KiCad E.D.A. 10.0.5 | Id: 6/13

06 Protected actuator branches and central DYNAMIXEL star injection

Each actuator has a separate protected VDD branch; U2D2 pin 2 and inter–actuator VDD paths are omitted.



NOTE 1: INJ1 is a fixed central star: U2D2 pin 2 is unrouted and J1/J2/J3 positive rails never join.

NOTE 2: Exact actuator cables, branch returns, enclosure, continuity, isolation, signal–integrity, pull and no–backfeed tests remain release gates.

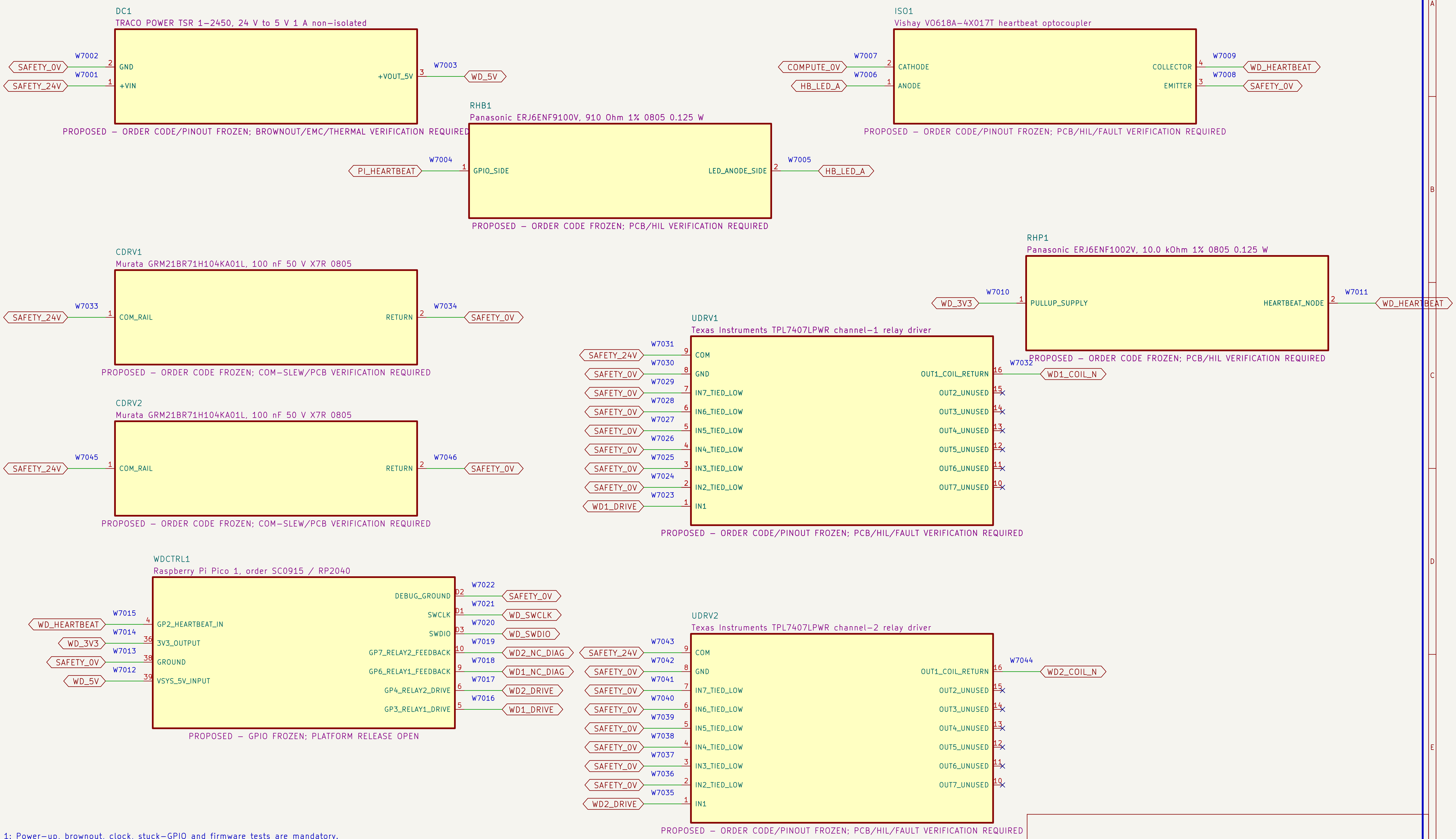
CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /06 Protected actuator branches and central DYNAMIXEL star injection/
File: 06_branches_and_injection.kicad_sch

Title: PB HR–V0 V3–P1.4 – 06

Size: A3 | Date: 2026–08–07 | **Rev: V3–P1.4**
KiCad E.D.A. 10.0.5 | Id: 7/13

07 Independent watchdog power, controller and drivers

The ordinary watchdog controller and drivers receive no safety–integrity credit by assertion.



NOTE 1: Power–up, brownout, clock, stuck–GPIO and firmware tests are mandatory.

NOTE 2: Qualified review decides whether watchdog loss is credited or diagnostic only;

current safety credit is NONE.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION

Project Button

Sheet: /07 Independent watchdog power, controller and drivers/
File: 07_watchdog_control.kicad_sch

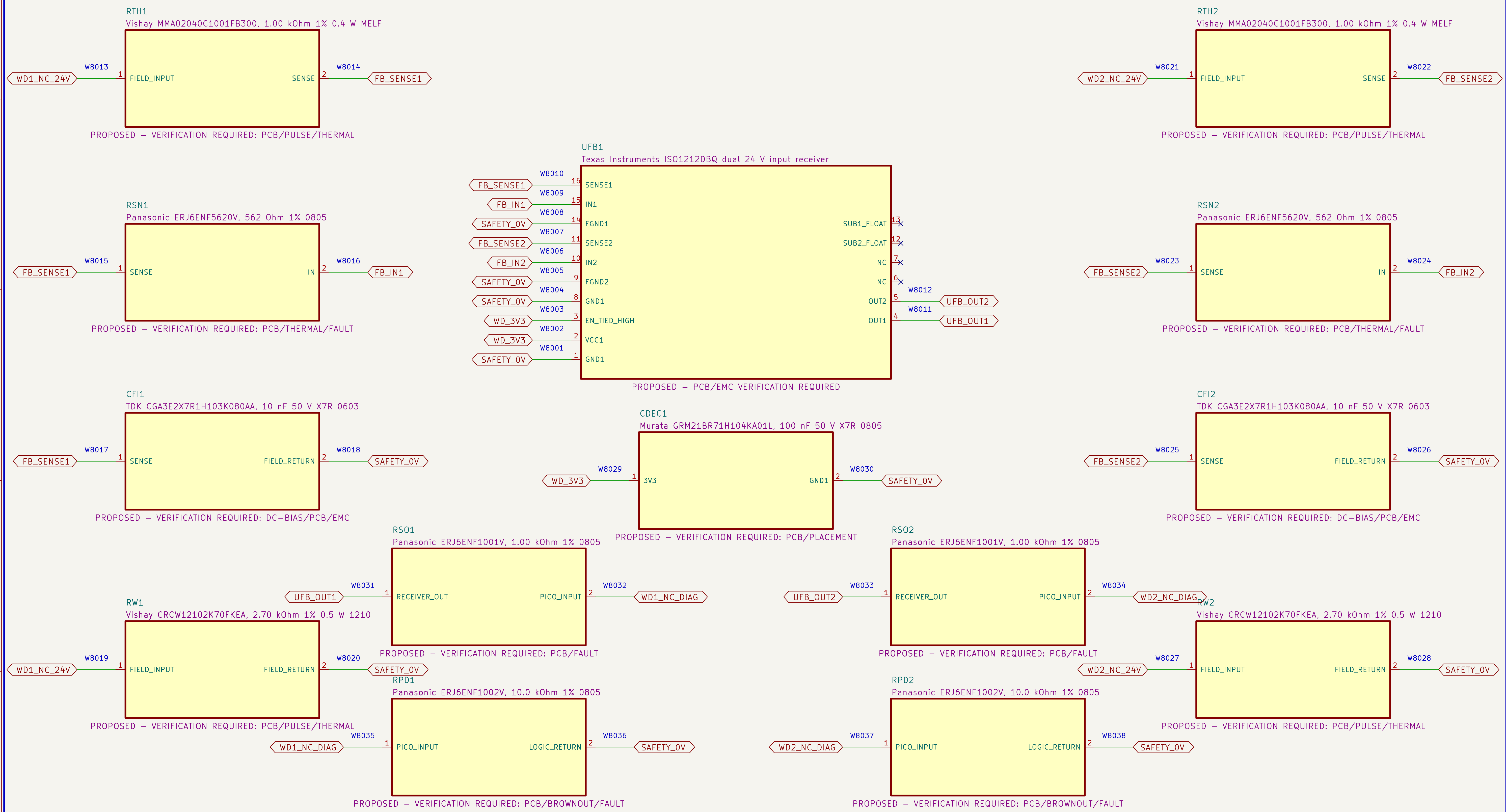
Title: PB HR–V0 V3–P1.4 – 07

Size: A3	Date: 2026–08–07	**Rev: V3–P1.4**
KiCad E.D.A. 10.0.5		Id: 8/13

PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION

08 Calculated dual-channel 24 V watchdog feedback

ISO1212DBQ converts both relay NC diagnostics to default-low 3.3 V logic; no isolation or safety credit is claimed.



NOTE 1: Type-3 values: RTHR=1.00 kOhm, RSENSE=562 Ohm, CIN=10 nF; calculated wetting load is 2.70 kOhm 1% 0.5 W per channel.

NOTE 2: Both grounds are SAFETY_0V; the ISO1212 barrier is not credited. Passive identities are frozen; PCB, derating, EMC and fault tests remain open.

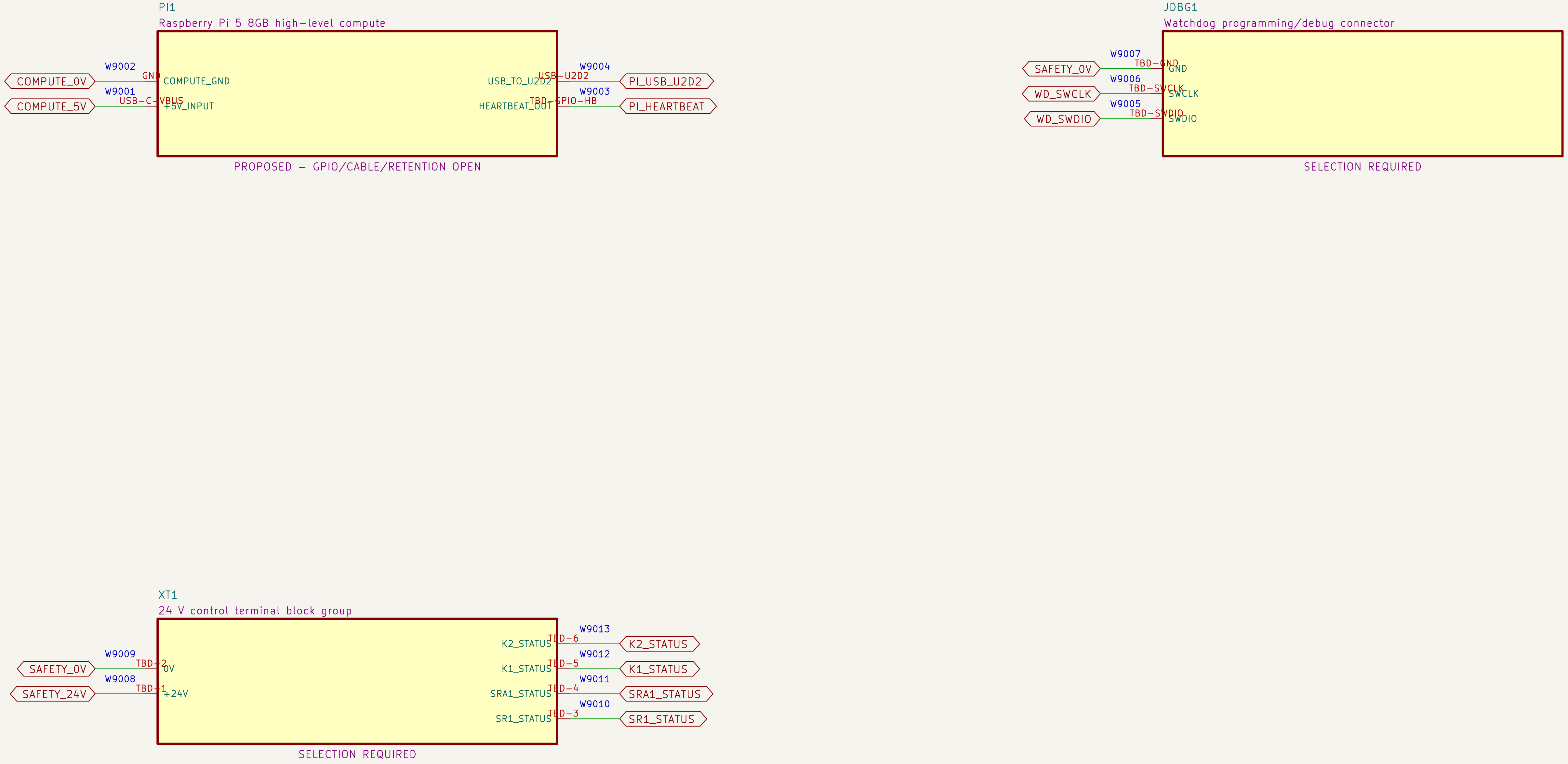
CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /08 Calculated dual-channel 24 V watchdog feedback/
File: 08_watchdog_feedback_interface.kicad_sch

Title: PB HR-V0 V3-P1.4 – 08

Size: A3 | Date: 2026-08-07 | **Rev: V3-P1.4**
KiCad E.D.A. 10.0.5 | Id: 9/13

09 Compute, debug and control terminals

High–level compute and diagnostic wiring have no authority to bypass or restore the safety chain.



NOTE 1: Programming/debug connections must not enable outputs during operation.

NOTE 2: Terminal family, markers, conductor range, torque and enclosure layout remain selection required.

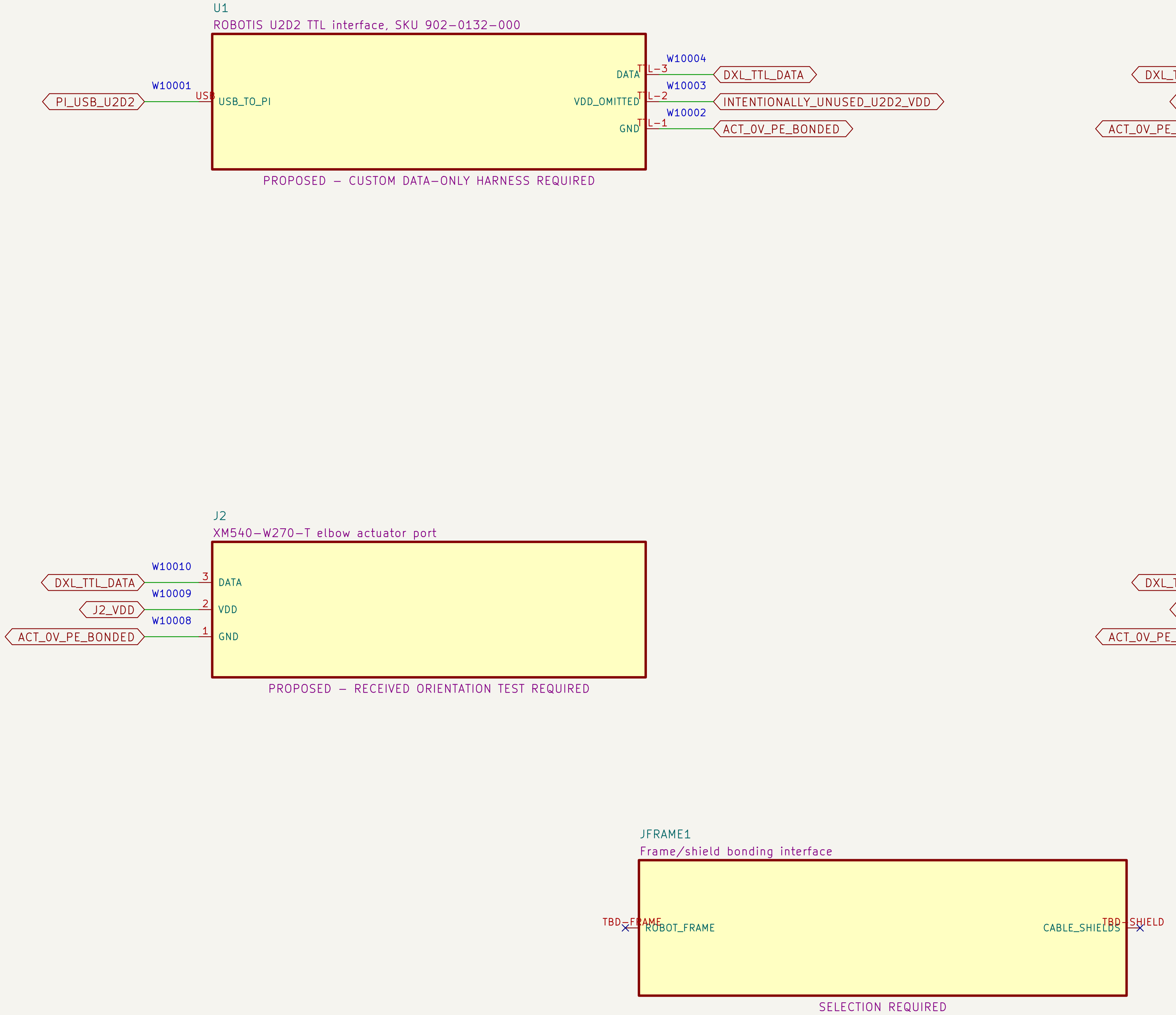
CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /09 Compute, debug and control terminals/
File: 09_compute_and_control_terminals.kicad_sch

Title: PB HR–V0 V3–P1.4 – 09

Size: A3	Date: 2026–08–07	**Rev: V3–P1.4**
KiCad E.D.A. 10.0.5		Id: 10/13

10 U2D2, actuator ports and bonding boundary

The U2D2 cable carries DATA and GND only; protected VDD is injected at each actuator.



NOTE 1: The V0 bus is TTL; HR-30 RS-485 remains a separate architecture.

NOTE 2: Do not bond frame or shields to 0V/PE until EMC and parallel-path review accepts the exact implementation.

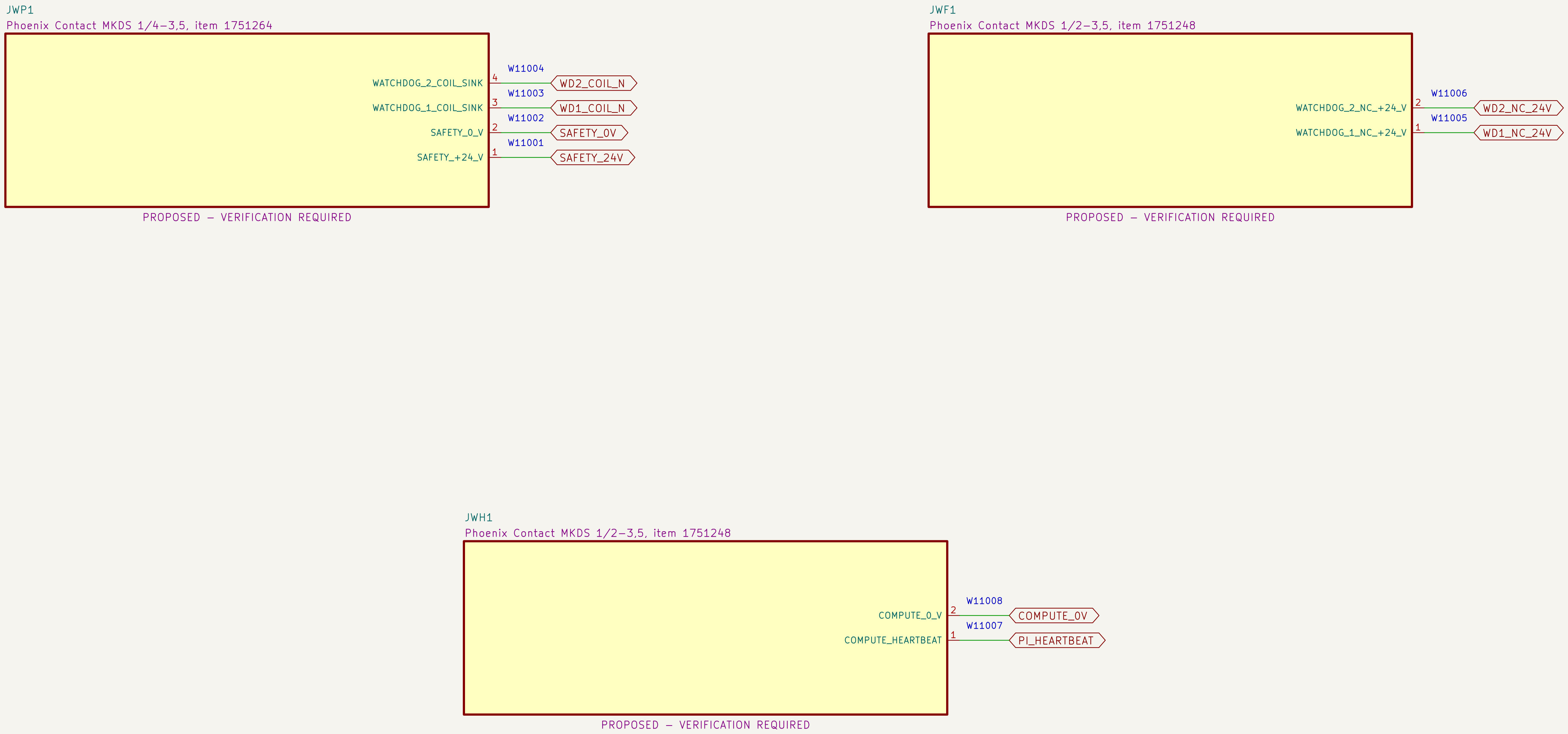
CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /10 U2D2, actuator ports and bonding boundary/
File: 10_actuator_interfaces.kicad_sch

Title: PB HR-V0 V3-P1.4 – 10

Size: A3	Date: 2026-08-07	**Rev: V3-P1.4**
KiCad E.D.A. 10.0.5		Id: 11/13

11 Watchdog PCB external connectors

Exact PCB terminal–block candidates and project pin allocations define the board boundary; harness release remains open.



NOTE 1: Terminal numbering follows the PCB footprint and must be confirmed against received parts before wiring.

NOTE 2: Nominal terminal ratings do not release branch protection, conductor, ferrule, harness, enclosure or thermal application.

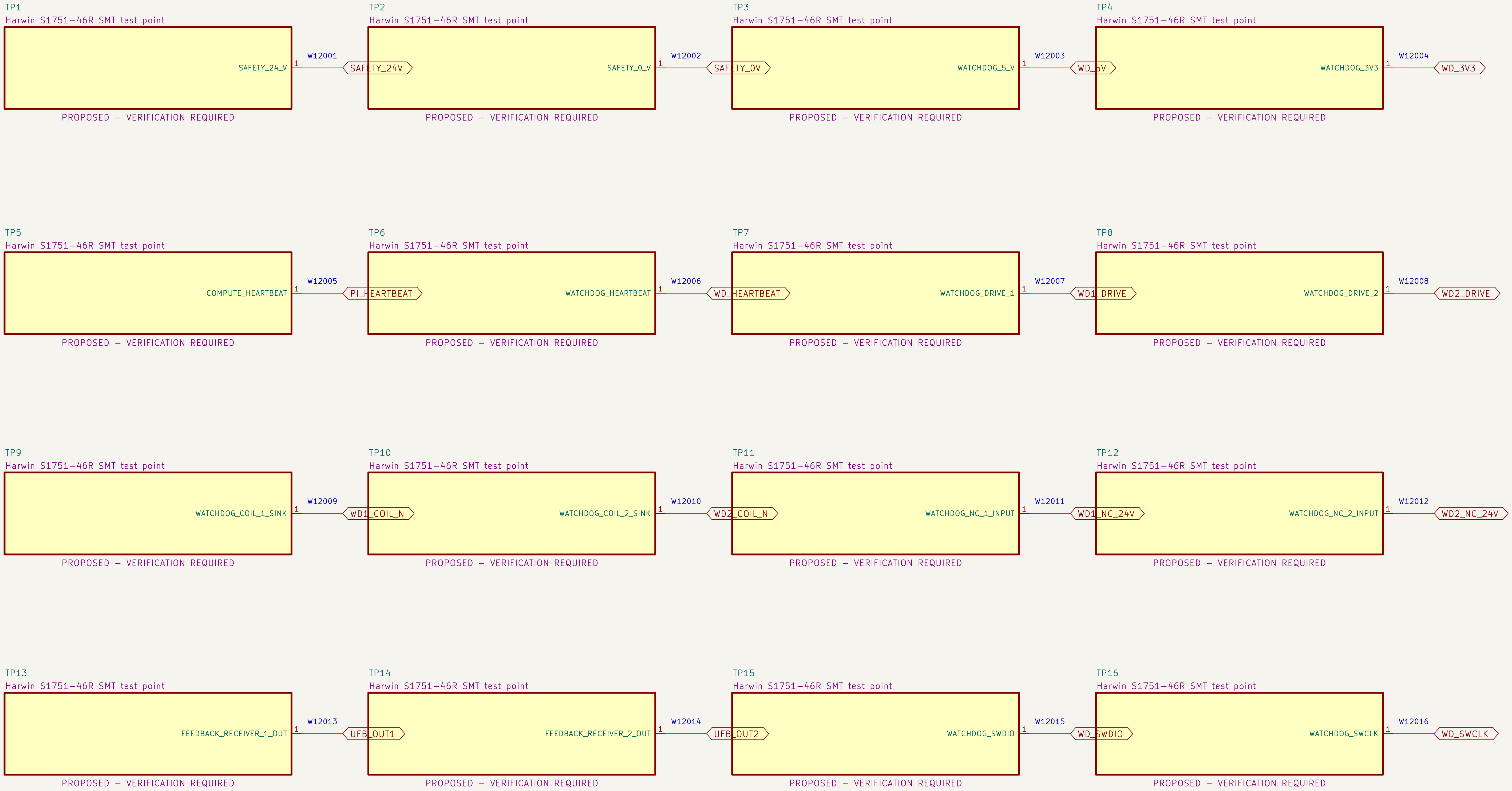
CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /11 Watchdog PCB external connectors/
File: 11_watchdog_pcb_connectors.kicad_sch

Title: PB HR-V0 V3-P1.4 – 11

Size: A3	Date: 2026-08-07	**Rev: V3-P1.4**
KiCad E.D.A. 10.0.5		Id: 12/13

12 Watchdog PCB test access

Exact clip–compatible test terminals expose controlled nodes without granting safety or operating authority.



NOTE 1: Test terminals are diagnostic features only; they provide no safety integrity or bypass authority.

NOTE 2: Verify clip clearance with guards, harnesses and power removed before any live measurement procedure is released.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /12 Watchdog PCB test access/
File: 12_watchdog_pcb_test_access.kicad_sch
Title: PB HR–V0 V3–P1.4 – 12

Size: A3	Date: 2026–08–07	**Rev: V3–P1.4**
KiCad E.D.A. 10.0.5		Id: 13/13